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2. NATURAL RESOURCE MANAGEMENT

2.1 WATER QUALITY LEP 2000 Cl 8(a)



Photograph - Drainage Structures on Ploughmans Creek

STORMWATER AND DRAINAGE ISSUES

Development that concentrates, redirects flows, increases flow rates or disturbs land in close proximity to creeks has the potential to affect waterways with associated erosion, sedimentation and release of nutrients, which combine to affect downstream water quality. Mapping of the Orange area has revealed streambank and gully erosion in various waterways of the City, particularly the downstream sections of Ploughmans Creek and tributaries of Summer Hill Creek. Erosion appears to be particularly prevalent within the "North Orange" red earth soil types. This soil type largely affects the Orange urban area.

Urban stormwater has also been identified as a significant source of water pollutants. Uncontrolled stormwater flows from urban development has the potential to contribute to downstream impacts including periodic flooding, erosion, sedimentation and reduced in-stream water quality.

Orange City Council's Drainage Strategy was adopted by Council on 19 June 1997 to identify measures required to control stormwater-flow velocities and volumes through the urban area, predominantly the Blackman's Swamp Creek Catchment.

A feasibility study for the construction of wetlands in a number of locations in and around the urban area established the potential for constructing artificial wetlands and gross pollutant traps (GPTs) in the City in conjunction with, and additional to, the detention basins in the drainage strategy.

P0 2.1-1 PLANNING OUTCOMES - STORMWATER QUALITY

1 Development is carried out in a manner that does not contribute to downstream erosion or sedimentation of waterways.
2 Development complies with the *Water and Soil Erosion Control* requirements of the *Development and Subdivision Code.*
3 On-site detention is carried out in accordance with the *Development and Subdivision Code* for all developments comprising buildings with a site coverage greater than 50m^2 or where site coverage exceeds the "percentage impervious" level listed in the Code applicable to that development.
4 Where on-site detention is not appropriate, contributions are made towards retarding basins and/or GPTs and associated drainage under the Contribution Plan that applies to the land.
5 Development in the vicinity of natural watercourses is positioned away from the waterway and includes measures to minimise the impact of the development on the waterway such as the establishment of creekside buffer zones and planting of native trees in a manner that enhances streambank stability.

RELATED INFORMATION – WATER QUALITY

1 Lyall and Macoun Consulting Engineers, *Orange Stormwater Drainage Strategy Study* unpublished report for Orange City Council.
2 Morse McVey & Associates. *Feasibility study for the construction of wetlands at nine locations in the City of Orange.* Unpublished study for Orange City Council
3 Orange City Council, *Development and Subdivision Code*

GROUNDWATER ISSUES

Groundwater has played a significant function in providing water for the City since its early development with a number of wells being established in the original Village of Orange. The village population, however, also experienced health effects in the late 19th Century as a result of pollution from septic tanks in proximity to these wells (Orange Heritage Study).

The 1987 Map from the former Department of Water Resources (now DLWC) explains potential pollution impact on groundwater as follows:

> *Groundwater is an important resource and although the risk of pollution is less than the risk to surface waters, the consequences if pollution does occur are severe because it may be impractical or impossible to rehabilitate a polluted aquifer. Some important possible pollutants are sewage effluent from treatment works and septic tanks, waste from manufacturing and vehicle service industries, fertilisers and pesticides.*

Development involving groundwater extraction and/or on-site wastewater disposal is deemed to have the potential to affect groundwater resources. Groundwater monitoring has been established for specific developments in the City including Council's Resource Recovery Centre, Orange General Cemetery, quarries and industrial on-site wastewater irrigation. Council may require that development with the potential to affect groundwater include information on potential risks and proposed management measures to protect groundwater resources.

PO 2.1-2 PLANNING OUTCOMES - GROUNDWATER QUALITY

1 Development applications for development (excluding dwelling houses) that proposes to extract groundwater or involve on-site wastewater disposal identify potential risks to, and management of, groundwater resources.
2 Development is carried out in a manner that does not adversely affect groundwater resources.
3 Development considered by Council to have the potential to significantly affect groundwater quality incorporates a monitoring program and provides test results from a NATA-accredited laboratory to Council for review and for inclusion in the City SoE Reports.

PO 2.1-2 PLANNING OUTCOMES - GROUNDWATER QUALITY

4 Development that requires or proposes the use of groundwater demonstrates that the groundwater extraction will meet the requirements of DLWC, where necessary.

RELATED INFORMATION – GROUNDWATER ISSUES

1 Department of Water Resources 1987 *Groundwater in New South Wales: Assessment of Pollution Risk*. (OCC Technical Services Map GM21:2)
2 Department of Local Government 1998 *Environment & Health Protection Guidelines: On-site Sewage management for Single Households*
3 DIPNR web page: Caring for our Natural Resources - Water

2.2 SOIL RESOURCES LEP 2000 Cl 8(a)

ISSUES

The primary soil groups of the City are derived from the volcanic geology of the area. Dominant soil landscapes of the City comprise fertile Kraznozems of the "Towac" and "Spring Hill" soil types on slopes over basalt flows in the south of the Orange LGA; and the North Orange Red Earths to the north and the east of the urban area.

The Towac and Spring Hill soil types have moderate inherent chemical fertility and permeability with low salinity, low erosion hazard and moderate shrink-swell potential. Soils commonly found in drainage depressions are lower in fertility and higher in shrink-swell potential.

The dominant North Orange Red Earths have low chemical fertility, moderate to high salinity, moderate erosion hazard, high permeability and low shrink-swell potential.

These soil characteristics influence land use and development capability. Soil characteristics determine the suitability for building footings, on-site waste disposal, road engineering and drainage. Shallow soils also affect construction and on-site wastewater disposal solutions.

PO 2.2-1 PLANNING OUTCOMES – SOIL RESOURCE MANAGEMENT

1 Development complies with the *Water and Soil Erosion Control* requirements of the *Development and Subdivision Code.*
2 Sites affected by soil degradation are restored in accordance with management strategies to be submitted with development proposals.
3 Agricultural practices apply conservation farming techniques particularly within the water supply catchments and in areas susceptible to significant erosion hazard.
4 A geotechnical investigation is carried out by a NATA-accredited laboratory that identifies and classifies all new residential lots for dwelling houses in accordance with AS 2870-1996 *Residential Slabs and Footings Construction*.
5 A geotechnical investigation is undertaken that determines the suitability of land for on-site disposal of sewage effluent in accordance with *Environmental and Health Protection Guidelines: On-site Sewage management for Single Households* where appropriate.
6 Non-agricultural activities in rural areas are carried out on less -productive soils.

RELATED INFORMATION – SOIL RESOURCES

1 Australian Standard AS1547 *Disposal systems for effluent from domestic premises*
2 Australian Standard AS 2870-1996 *Residential Slabs and Footings Construction*.
3 Department of Local Government 1998 *Environment & Health Protection Guidelines: On-site Sewage Management for Single Households*
4 DIPNR web page: Caring for our natural resources – Soil

2.3 VEGETATION LEP 2000 Cl 8(a)



Photograph - Remnant bushland at Clifton Grove

From figures derived from NPWS mapping of remnant vegetation, as recorded in *Orange State of Environment* Reports, native vegetation comprises less than 9% of the area of rural land in the City - of which the majority (8.3% of all rural land) is recognised to comprise disturbed regrowth forest.

The Department of Land and Water Conservation (DLWC) summarise the function of, and threats to, native vegetation as follows:

> *Native vegetation provides vital habitat for birds and animals. It can also provide direct economic benefits to landholders, such as sustainable timber management and food products, and indirect economic benefits such as shelter and shade for stock, crops and pasture.*
>
> *Our vegetation is under intense pressure as our population grows and the need for development continues. Loss of vegetation can lead to salinity, erosion, weeds, degradation and the loss of habitat for native flora and fauna* (DLWC, http://www.dlwc.nsw.gov.au/ care/veg/index.html).

The *Native Vegetation Conservation Act 1997* (NVC Act) has controlled the clearing of native vegetation across NSW since 1 January 1998. The NVC Act requires the approval for clearing of native vegetation unless the land or clearing is excluded from the Act or clearing is exempt from the need to obtain development consent. The Act is administered by the Department of Infrastructure, Planning and Natural Resources (formerly DLWC).
http://www.dlwc.nsw.gov.au/care/veg/index.html)

The heavily-modified landscape and climate of Orange have culminated in introduced species of trees providing a significant contribution to the landscape character. Established urban areas are characterised by “cottage” gardens in residential areas and by formal parks and avenues of exotic street trees. Native trees combine with introduced species primarily in parks forming creek corridors and in more recent suburban areas.

In addition to some patches of remnant native vegetation, rural areas comprise formal shelter belts, windbreaks and hedges predominantly of pine and hawthorns (*crataegous spp*) and gardens with large trees in parkland settings around homestead buildings, which complement the landscape character of orchards and vineyards. Remnant native vegetation is scattered in isolated areas on rural land and within road reserves.

PO 2.3-1 PLANNING OUTCOMES – VEGETATION MANAGEMENT

1 Compliance with the *Native Vegetation Conservation Act 1997*.
2 Development is designed and constructed in a way that minimises the impact on existing vegetation.
3 Particular attention is given to the effect of rural or urban residential release development on existing vegetation and scenic areas.
4 Development applications indicate on plans the location of all significant trees affected by or in the vicinity of development.
5 Applications demonstrate to Council's satisfaction that all practical measures have been made to retain trees that contribute to and embellish the Orange landscape.

RELATED INFORMATION - VEGETATION
Land and Water Australia research: The value of native vegetation: *Urban and rural perspectives;*
DIPNR webpage: Caring for our natural resources - vegetation

2.4 FLORA AND FAUNA, BIODIVERSITY
LEP 2000 Cl 8(a)

The natural environment of the Orange LGA has been heavily modified as a consequence of land clearing for various uses including agriculture, plantation forests, mining and urban development. The clearing of native vegetation has significantly affected native habitats.

The *Threatened Species Conservation Act 1995* requires that a *Species Impact Statement* is prepared and submitted with an application for development on land that is all or part of a "Critical Habitat" or that is likely to affect threatened species, populations or ecological communities, or their habitats.

At the time that this Plan took effect, there were no "critical habitats" declared within the Orange LGA. The NPWS Atlas of NSW Wildlife contains a single listing of a threatened species, the Blue-billed Duck (*Oxyura australis*). Blue-billed and Freckled Ducks (*Stictonetta naevosa*) have also been recorded at Spring Creek Reservoir.

The Mount Canobolas *Xanthoparmelia* Lichen community has been declared to be an endangered ecological community under Part 3 of Schedule 1 to the *Threatened Species Conservation Act 1995*. The lichen community may be found on and adjacent to The Pinnacle on the western boundary of the City

White Box, Yellow Box Blakely's Red Gum Woodland is an endangered ecological community under the TSC Act and is known to occur in the Orange region.

The Commonwealth *Environment Protection and Biodiversity Conservation Act 1999* commenced on 16 July 2000. Development significantly affecting nationally-listed threatened species and ecological communities will require approval under that Act.

PO 2.4-1 PLANNING OUTCOMES - FLORA & FAUNA MANAGEMENT

1 Where there is a likely impact of development on native habitats, that impact is addressed in the development application.
2 A *Species Impact Statement* is prepared for development that is likely to significantly affect habitats of threatened species. The statement is submitted with a development application and indicates how threatened species will be managed with the development.
3 Development affecting all or part of significant water bodies or remnant woodland areas with the potential to comprise habitats of threatened species incorporates the protection and conservation of these areas where deemed reasonable by Council.
4 Threatened species, populations and ecological communities are managed in conjunction with development in accordance with the *Threatened Species Conservation Act*.

RELATED INFORMATION – FLORA AND FAUNA, BIODIVERSITY

1 Orange City Council. ***State of Environment Reports 1993 to 1996/97***.
2 National Parks and Wildlife Service ***Threatened Species Management Information Circulars,*** 1996 NPWS.
3 Beukers, Max, 1996 ***Options for Conserving Biodiversity in Bathurst District***, NPWS
4 NPWS native plants and animals listings

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